

```
%-----SCRIPT 1-----
```

```
Caf = 40;

V=linspace(0,0.1,100);
v=1;
vr=10;

CaGuess = 0;
CbGuess = 0;
CcGuess = 0;
C(100,1) = 100;
C(100,2) = 100;
C(100,3) = 100;

while abs(CaGuess-C(100,1))>0.01
    while abs(CbGuess-C(100,2))>0.01
        while abs(CcGuess-C(100,3))>0.01

            CaGuess = C(100,1);
            CbGuess = C(100,2);
            CcGuess = C(100,3);

            CaBound = (v*Caf+vr*CaGuess)/(v+vr);
            CbBound = (vr*CbGuess)/(v+vr);
            CcBound = (vr*CcGuess)/(v+vr);

            [V,C] = ode45(@ODEs,V, [CaBound, CbBound, CcBound]);

        end
    end
end

Selectivity = C(100,1)/Caf;
```

```
%-----SCRIPT 2-----
```

```
vr=[0 0.1 0.5 1];

Selectivity = [0.998 0.9978 0.9970 0.9961];%%obtained via iteration.
plot(vr, Selectivity);
hold on;
xlabel('Recycle flow, L/s');
ylabel('Conversion');
title('Conversion vs Recycle');
%%code does not converge at vr >10 without resulting in selectivity >1.
%%Code needs time to process completely.
%%plot of only converged data shown.
```

```
%-----SCRIPT 3-----
```

```
function dC = ODEs(t,C)
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```
k(1)=0.001;
```

```
k(2)=0.0001;
```

```
Caf = 40;
```

```
L=10;
```

```
A=0.01;
```

```
v=1;
```

```
vr=1;
```

```
dC(1,1) = -k(1)*C(1)^2/(v+vr);
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```
dC(2,1) = (k(1)*C(1)^2-k(2)*C(2))/(v+vr);
```

```
dC(3,1) = k(2)*C(2)/(v+vr);
```

```
end
```